① If $u = x^2y + y^2z + z^2x$, prove that

$$\frac{\partial u}{\partial x} + \frac{\partial u}{\partial y} + \frac{\partial u}{\partial z} = (x+y+z)^2.$$

② If $z = e^{(ax+by)} f(ax-by)$, prove that

$$b\frac{\partial z}{\partial x} + a\frac{\partial z}{\partial y} = 2abz.$$

③ If $z(x+y) = x^2+y^2$, show that

$$\left(\frac{\partial z}{\partial x} - \frac{\partial z}{\partial y}\right)^2 = 4\left[1 - \frac{\partial z}{\partial x} - \frac{\partial z}{\partial y}\right]$$

④ If $\theta = t^n e^{-r^2/4t}$, find what value of n will make

$$\frac{1}{r^2}\frac{\partial}{\partial r}\left(r^2\frac{\partial \theta}{\partial r}\right) = \frac{\partial \theta}{\partial t}.$$

⑤ If $u = \log(x^3+y^3+z^3-3xyz)$ show that

(a) $\frac{\partial u}{\partial x} + \frac{\partial u}{\partial y} + \frac{\partial u}{\partial z} = \frac{3}{x+y+z}$

(b) $\left(\frac{\partial}{\partial x} + \frac{\partial}{\partial y} + \frac{\partial}{\partial z}\right)^2 u = -\frac{9}{(x+y+z)^2}$

⑥ If $u = \log\left(\frac{x^2+y^2}{\sqrt{x}+\sqrt{y}}\right)$, find the value of

(i) $x\frac{\partial u}{\partial x} + y\frac{\partial u}{\partial y}$ (ii) $x^2\frac{\partial^2 u}{\partial x^2} + 2xy\frac{\partial^2 u}{\partial x \partial y} + y^2\frac{\partial^2 u}{\partial y^2}$.

⑦ If $u = \log\left(\frac{x^4-y^4}{x-y}\right)$, prove that

(i) $x\frac{\partial u}{\partial x} + y\frac{\partial u}{\partial y} = 3$ (ii) $x^2\frac{\partial^2 u}{\partial x^2} + 2xy\frac{\partial^2 u}{\partial x \partial y} + y^2\frac{\partial^2 u}{\partial y^2} = -3$

⑧ If $u = (x^2+y^2)^{1/3}$, show that

$$x^2\frac{\partial^2 u}{\partial x^2} + 2xy\frac{\partial^2 u}{\partial x \partial y} + y^2\frac{\partial^2 u}{\partial y^2} = -\frac{2}{9}u$$

(9) If $u = \tan^{-1}\left[\frac{x^{1/3}+y^{1/3}}{x^{1/3}-y^{1/3}}\right]$, find the value of

i) $x\frac{\partial u}{\partial x} + y\frac{\partial u}{\partial y}$ (ii) $x^2\frac{\partial^2 u}{\partial x^2} + 2xy\frac{\partial^2 u}{\partial x \partial y} + y^2\frac{\partial^2 u}{\partial y^2}$.

(10) If $u = \tan^{-1}\left(\frac{x^3+y^3}{\sqrt{x}+\sqrt{y}}\right)$, find the value of

i) $x\frac{\partial u}{\partial x} + y\frac{\partial u}{\partial y}$ (ii) $x^2\frac{\partial^2 u}{\partial x^2} + 2xy\frac{\partial^2 u}{\partial x \partial y} + y^2\frac{\partial^2 u}{\partial y^2}$.

(11) If $u = f(x-y, y-z, z-x)$, prove that

$$\frac{\partial u}{\partial x} + \frac{\partial u}{\partial y} + \frac{\partial u}{\partial z} = 0$$

(12) If $u = f(2x-3y, 3y-4z, 4z-2x)$, prove that

$$\frac{1}{2}\frac{\partial u}{\partial x} + \frac{1}{3}\frac{\partial u}{\partial y} + \frac{1}{4}\frac{\partial u}{\partial z} = 0$$

(13) If $\phi\left(\frac{z}{x^3}, \frac{y}{x}\right) = 0$ then show that $xp + yq = 3z$

(14) If $u = f\left(\frac{y-z}{xy}, \frac{z-x}{xz}\right)$, find the value of

$$x^2\frac{\partial u}{\partial x} + y^2\frac{\partial u}{\partial y} + z^2\frac{\partial u}{\partial z}.$$

(15) If $u = f(x^2+2yz, y^2+2xz)$, find the value of

$$(y^2-xz)\frac{\partial u}{\partial x} + (x^2-yz)\frac{\partial u}{\partial y} + (z^2-xy)\frac{\partial u}{\partial z}.$$

(16) If $u = \frac{yz}{x}$, $v = \frac{xz}{y}$, $w = \frac{xy}{z}$, find $\frac{\partial(x,y,z)}{\partial(u,v,w)}$

(17) If $u = x+y+z$, $uv = y+z$, $uvw = z$

show that $\frac{\partial(u,v,w)}{\partial(x,y,z)} = u^3$

(18) If $u = 3x+2y-z$, $v = x-2y+z$, $w = x(x+2y-z)$, functionally related and find relation

(19) If $u = x+y+z$, $v = x^2+y^2+z^2$, $w = x^3+y^3+z^3-3xyz$ Are u, v, and w functionally related? If so, find relationship.

(20) If $u = \frac{x+y}{1-xy}$ and $v = \tan^{-1}x + \tan^{-1}y$, find $\frac{\partial(u,v)}{\partial(x,y)}$. Are u and v functionally related? If so, find relationship.